

A Demonstration of Selfrionette: A Force-Input Controller for Continuous Full-Body Avatar Manipulation and Enhanced Virtual Haptics

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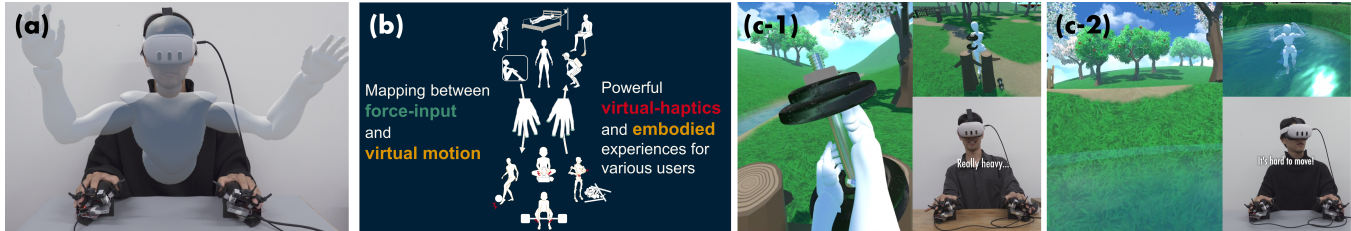


Figure 1: (a) Selfrionette. (b) Concept diagram of Selfrionette. It converts the force input from the user's fingers into the movement of a virtual avatar, enabling a more refined embodiment and haptic experience. (c-1) Demonstration Scene 1: Users can move a virtual hand and grasp objects. The force required to move the hand varies depending on the weight of the object. (c-2) Demonstration Scene 2: Users can freely navigate through the virtual space. The force input required for walking also varies depending on changes in environmental characteristics, such as walking underwater.

Abstract

This paper proposes a grasped-type controller that utilizes user's fingers' force-inputs into an embodied and continuous virtual self body motion in virtual reality. The proposed system allows users embodied interactions with virtual objects and walk in VR only by their physical fingers' force-inputs, and therefore, it has a potential to overcome users' physical and spatial constraints of a physical environment. Moreover, the force-input can present a reaction force to the user's finger. The proposed system takes into account the physical property of interacting objects and adjusts relationship between force input and virtual body motion to achieve realistic haptic interaction.

CCS Concepts

• **Hardware** → **Emerging interfaces**; • **Human-centered computing** → **Virtual reality**; *Accessibility technologies*; *Haptic devices*.

Keywords

Virtual Body Manipulation, Avatar Manipulation, Pseudo-haptics, Embodiment, Body Schema

*Both authors contributed equally to this research.

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1 Introduction

The concept of embodiment-where users feel as if the avatar is their own body-is one of the most interesting topic in the field of virtual reality (VR) studies because it significantly affects the realism, immersion, and presence within virtual environments. The majority of embodiment research has concentrated on creating a morphological one-to-one correspondence between the user's body and the avatar, with a particular focus on motion tracking techniques. These techniques allow users to apply their familiar body control schemas facilitating an easier transition to experiencing embodiment with the avatar. However, it has the challenge that users with spatial limitations or physical disabilities may not be able to take full advantage of it. Furthermore, another key challenge of the virtual full body interactions is haptic feedback. If we want to realize full body haptic interactions, users need to equip bulky devices [Hashimoto et al. 2022], leading to the constraints of users' motions or low usability. Despite the evident environmental and physical constraints associated with avatar manipulation interfaces, research into achieving a high sense of embodiment in VR through alternative interfaces has been limited. In response to these challenges, this study proposes a novel system that transforms fingers' force-input into continuous virtual body movements. This system aims to enable embodied and continuous virtual body manipulation, overcoming the physical and spatial constraints of users, and

to provide an enhanced virtual haptic experiences. This system, called Selfrionette (Self + Marionette), marks a significant advance in creating more immersive and realistic virtual experiences.

2 Selfrionette

2.1 Rationale behind Selfrionette

In this section, we discuss the rationale behind Selfrionette. The first critical inquiry arises whether a system like Selfrionette, which entails morphologically distinct input-output correspondences in avatar manipulation, can update the body schema and elicit a sense of embodiment. In the field of prosthetic research, it has been observed that users can perceive visual and haptic stimuli as congruent information, even when the locations of visual presentation (e.g., the fingertips of a prosthetic hand) and haptic stimulation (e.g., the residual limb) differ, as long as the expected haptic sensation matches the actual sensation presented (modality matching) [Svensson et al. 2017]. This reduces cognitive effort required for users to acclimate to prosthetic control and enhances the sense of embodiment. Then, the question is if Selfrionette system can present modality-matched feedback. The expected feedback while virtual body movements and interactions are variations in muscle and tendon sensations (proprioceptive information) and/or skin deformation (cutaneous information). While, Selfrionette employs force-input, wherein the force exerted by the user on a sensor triggers an equivalent reactive force, which in turn, leads to alterations in the proprioceptive information and skin deformation of the fingers. Therefore, Selfrionette potentially offers modality-matched feedback during VR interactions. Moreover, we posit that by taking into account the physical property of interacting objects and adjusting relationship between force input and virtual body motion, Selfrionette could present virtual haptics. Indeed, our previous study suggests that this method can convey multiple levels of perceived weights during virtual object lifting in VR [Hirao et al. 2023]. Based on these considerations, we believe Selfrionette holds the potential to enable embodied virtual body manipulation that overcomes physical and spatial constraints and to extend the capabilities of virtual haptics presentation.

2.2 Implementation

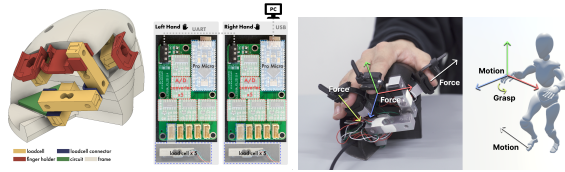


Figure 2: Left) Components. Figure 3: Correspondence Right) System configura-tion.

Figure 2 shows the appearance and components of the device, as well as the system diagram. The hardware for one hand consists of a total of five single-beam load cells and a spherical casing where users can place their hand in a natural posture. The five load cells are allocated with one for the thumb, one for the middle finger,

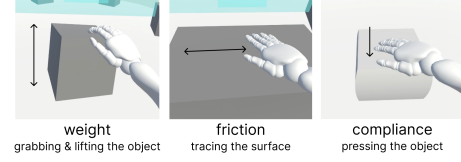


Figure 4: Interaction with objects of each haptic property.

and three for the index finger in an orthogonal arrangement. By connecting two devices, a total of ten degrees of freedom input can be acquired for both hands. Figure 3 shows the correspondence between force inputs from the device and avatar movements. The values from the three load cells on the index finger correspond to the movements along the respective coordinate axes related to the position of the virtual hands. The value from the thumb's load cell is used to control the lifting and lowering movements of the virtual legs, and locomotion in VR. Meanwhile, the value from the middle finger's load cell is used as a trigger for hand interactions, such as grasping and releasing virtual objects. For the legs and hands, the force applied to each load cell is converted into a force acting on the corresponding target point in VR space. In this study, stable motion was achieved by defining the behavior as shown in Equation 1. Specifically, the force applied to the load cell F_l is proportional to the force applied to the target in VR F_v by the proportionality constant α , and the behavior of the motion of each target $\ddot{x}$ is described by a spring-damper model. More detailed information is provided in our previous paper [Hashimoto and Hirao 2024].

$$F_v = \alpha F_l, \quad m_v \ddot{x} + c \dot{x} + k(x - x_0) = F_v + f_m \quad (1)$$

3 Demonstration at SIGGRAPH Asia

During the demonstration at the venue, participants can experience freely moving their avatar and feel the differences in haptic perceptions when interacting with virtual objects. Diverse haptic perceptions can be achieved by changing f_m in Equation 1, such that simulating virtual weight as $f_m = -m_{obj} \ddot{x} - m_{obj} g$, virtual viscosity as $f_m = -c \dot{x}$ and virtual compliance as $f_m = -k_{obj} x$. For the haptic experience, we prepare three types of haptic objects as shown in Figure 4. Each object will represent differences in weight, friction, and compliance, allowing participants to clearly perceive the variations in these haptic properties.

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